

7(a)(i)	(proton is uud so) $(2/3)e + (2/3)e - (1/3)e = e$	B1									
7(a)(ii)	(neutron is udd so) $(2/3)e - (1/3)e - (1/3)e = 0$	B1									
7(b)(i)	<table border="1"> <tr> <td></td><td>β^-</td><td>β^+</td></tr> <tr> <td>nucleon number</td><td>90</td><td>64</td></tr> <tr> <td>proton number</td><td>39</td><td>28</td></tr> </table> <i>all correct</i>		β^-	β^+	nucleon number	90	64	proton number	39	28	B1
	β^-	β^+									
nucleon number	90	64									
proton number	39	28									
7(b)(ii)	weak (nuclear force/interaction)	B1									
7(b)(iii)	β^- decay: electron and (electron) antineutrino β^+ decay: positron and (electron) neutrino <i>all correct</i>	B1									